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# AP<sup>®</sup> Calculus BC

## Sample Student Responses and Scoring Commentary

### **Inside:**

#### **Free-Response Question 5**

- ☒ **Scoring Guidelines**
- ☒ **Student Samples**
- ☒ **Scoring Commentary**

**Part B (BC): Graphing calculator not allowed****Question 5****9 points****General Scoring Notes**

- The model solution is presented using standard mathematical notation.
- Answers (numeric or algebraic) need not be simplified. Answers given as a decimal approximation should be accurate to three places after the decimal point. Within each individual free-response question, at most one point is not earned for inappropriate rounding.

Let  $y = f(x)$  be the particular solution to the differential equation  $\frac{dy}{dx} = (3 - x)y^2$  with initial condition  $f(1) = -1$ .

|          | Model Solution                                                                                                                                                             | Scoring      |                     |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------|
| <b>A</b> | Find $f''(1)$ , the value of $\frac{d^2y}{dx^2}$ at the point $(1, -1)$ . Show the work that leads to your answer.                                                         |              |                     |
|          | $\frac{d^2y}{dx^2} = -y^2 + (3 - x)2y\frac{dy}{dx}$                                                                                                                        | Product rule | <b>Point 1 (P1)</b> |
|          |                                                                                                                                                                            | Chain rule   | <b>Point 2 (P2)</b> |
|          | $f'(1) = \left. \frac{dy}{dx} \right _{(x,y)=(1,-1)} = (3 - 1)(-1)^2 = 2$<br>$f''(1) = \left. \frac{d^2y}{dx^2} \right _{(x,y)=(1,-1)} = -(-1)^2 + (3 - 1)(2)(-1)(2) = -9$ | $f''(1)$     | <b>Point 3 (P3)</b> |

## Scoring Notes for Part A

- The expression  $\frac{d^2y}{dx^2} = -y^2 + (3-x)2y$  or  $\frac{d^2y}{dx^2} = 6y - y^2 - 2xy$  earns **P1** but not **P2**. Such a response is not eligible for **P3**.
- The expression  $\frac{d^2y}{dx^2} = -2y\frac{dy}{dx}$  or  $\frac{d^2y}{dx^2} = 6y\left(\frac{dy}{dx}\right) - 2y\left(\frac{dy}{dx}\right)$  earns **P2** but not **P1**. Such a response is eligible for **P3** for a consistent answer of  $f''(1) = 4$  or  $f''(1) = -8$ , respectively, which is found by correctly substituting correct values for  $x$ ,  $y$ , and  $\frac{dy}{dx}$ .
- A response of  $-(-1)^2 + (3-1)(2)(-1)(2)$  earns **P1**, **P2**, and **P3** regardless of any subsequent errors in simplification.
- Alternate approach (using separation of variables):

The particular solution for the differential equation that passes through the point  $(1, -1)$  is

$$y = \frac{2}{x^2 - 6x + 3}. \text{ Therefore, } \frac{dy}{dx} = \frac{-2(2x - 6)}{(x^2 - 6x + 3)^2}.$$

This response has not yet earned **P1**, **P2**, or **P3**.

- A response that correctly applies the quotient rule (or product rule) and the chain rule to find that  $\frac{d^2y}{dx^2} = \frac{-4(x^2 - 6x + 3)^2 + 4(2x - 6)(x^2 - 6x + 3)(2x - 6)}{(x^2 - 6x + 3)^4}$  earns **P1** and **P2** and is eligible to earn **P3** for the correct answer of  $f''(1) = -9$ .
- A response that correctly applies the quotient rule (or product rule) but does not correctly apply the chain rule (e.g.,  $\frac{d^2y}{dx^2} = \frac{-4(x^2 - 6x + 3)^2 + 4(2x - 6)(x^2 - 6x + 3)}{(x^2 - 6x + 3)^4}$ ) earns **P1**, does not earn **P2**, and is not eligible to earn **P3**.
- A response that does not correctly apply the quotient rule (or product rule) but does correctly apply the chain rule (e.g.,  $\frac{d^2y}{dx^2} = \frac{-4}{2(x^2 - 6x + 3)(2x - 6)}$ ) does not earn **P1**, earns **P2**, and is eligible to earn **P3** for a consistent answer.

**B** Write the second-degree Taylor polynomial for  $f$  about  $x = 1$ .

$$f'(1) = 2 \text{ and } f''(1) = -9$$

Two terms

**Point 4 (P4)**

$$P_2(x) = -1 + 2(x - 1) - \frac{9}{2}(x - 1)^2$$

Remaining term

**Point 5 (P5)**

### Scoring Notes for Part B

- **P4** and **P5** can be earned with an answer consistent with incorrect values of  $f'(1)$  and  $f''(1)$  imported from part A.
- Any terms of degree greater than two or “+...” does not earn **P5**.
- A response of  $-1 + 2(x - 1) - \frac{9}{2}(x - 1)^2$  earns **P4** and **P5**, regardless of any subsequent algebraic simplification.
- A response that does not present the polynomial as powers of  $(x - 1)$  but instead presents a correct expanded/simplified form of the polynomial (e.g.,  $-\frac{9}{2}x^2 + 11x - \frac{15}{2}$ ) earns **P4** but not **P5**.

**C** The second-degree Taylor polynomial for  $f$  about  $x = 1$  is used to approximate  $f(1.1)$ . Given that  $|f'''(x)| \leq 60$  for all  $x$  in the interval  $1 \leq x \leq 1.1$ , use the Lagrange error bound to show that this approximation differs from  $f(1.1)$  by at most 0.01.

$$|f(1.1) - P_2(1.1)| \leq \frac{\max_{1 \leq x \leq 1.1} |f'''(x)|}{3!} |1.1 - 1|^3 \leq \frac{60}{6} (0.1)^3 = 0.01$$

Form of error bound

**Point 6 (P6)**

Analysis

**Point 7 (P7)**

### Scoring Notes for Part C

- **P6** is earned for presenting either  $\frac{\max_{1 \leq x \leq 1.1} |f'''(x)|}{3!} |1.1 - 1|^3$  or  $\frac{60}{6} (0.1)^3$ . Subsequent errors in simplification will not earn **P7**.
- To earn **P7**, a response must have earned **P6** and must explicitly connect the error bound with 0.01; for example by communicating Error  $\leq 0.01$ , Error Bound = 0.01, or equivalent.
- A response that declares the error is equal to 0.01 (or any equivalent form of this value) does not earn **P7**.

- D** Use Euler’s method, starting at  $x = 1$  with two steps of equal size, to approximate  $f(1.4)$ . Show the work that leads to your answer.

$$f(1.2) \approx f(1) + (1.2 - 1) \cdot \left. \frac{dy}{dx} \right|_{(1, -1)}$$

$$= -1 + 0.2(2) = -0.6$$

$$f(1.4) \approx f(1.2) + (1.4 - 1.2) \cdot \left. \frac{dy}{dx} \right|_{(1.2, -0.6)}$$

$$\approx -0.6 + 0.2(3 - 1.2)(-0.6)^2 = -0.4704$$

An approximation of  $f(1.4)$  is  $-0.47$ .

First step of  
Euler’s method

**Point 8 (P8)**

Answer with  
supporting work

**Point 9 (P9)**

#### Scoring Notes for Part D

- To earn **P8**, a response must demonstrate the first step of Euler’s method, with the correct initial condition, correct step size, and correct (or imported) expression for the derivative.  
Note: Any subsequent error in simplification or rounding will not affect the scoring for **P8**.
- The two steps of Euler’s method may be explicit expressions or may be presented in a table. For example:

| $x$ | $y$     | $\frac{dy}{dx} \cdot \Delta x$ (or $\frac{dy}{dx} \cdot 0.2$ ) |
|-----|---------|----------------------------------------------------------------|
| 1   | -1      | 0.4                                                            |
| 1.2 | -0.6    | 0.1296                                                         |
| 1.4 | -0.4704 |                                                                |

Note: In the presence of a correct answer, a table does not need to be labeled to earn both **P8** and **P9**. In the presence of no answer or an incorrect answer, such a table must be correctly labeled to earn **P8**.

- A response of  $-0.6 + 0.2(3 - 1.2)(-0.6)^2$  earns **P9**, regardless of any subsequent errors in simplification or rounding.
- A response that imports an incorrect value for  $f'(1)$  from part A or part B is eligible to earn **P9** with a consistent answer.
- A response may report the final answer as  $-0.47$ ,  $(1.4, -0.47)$ ,  $-\frac{294}{625}$ , or equivalent.

Q5



NO CALCULATOR ALLOWED

Q5

Answer QUESTION 5 PARTS A and B on this page.

PART A

$$f''(x) = \frac{d^2 y}{dx^2} = \frac{d}{dx}[(3-x)y^2] = -y^2 + 2(3-x)y \frac{dy}{dx} = 2(3-x)y^2 - y^2$$

$$f''(1) = 2(2)^2(-1)^3 - (-1)^2$$

PART B

$$T_2(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2}(x-1)^2 = -1 + 2(x-1) - \frac{9}{2}(x-1)^2$$

$$f'(1) = (3-1)(-1)^2 = 2$$

Page 12

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Q5



NO CALCULATOR ALLOWED

Q5

Answer QUESTION 5 PARTS C and D on this page.

PART C

$$|f(1.1) - T_2(1.1)| \leq \left| \frac{60}{3!} (1.1-1)^3 \right| \leq 0.01$$

PART D

$$\Delta x = \frac{1.4-1}{2} = 0.2$$

$$f(1.2) \approx -1 + 0.2(3-1)(-1)^2 = -0.6$$

$$f(1.4) \approx -0.6 + 0.2(3-1.2)(-0.6)^2$$

Q5

**NO CALCULATOR ALLOWED**

Q5

Answer QUESTION 5 PARTS A and B on this page.

PART A

$$f''(x) = \frac{d^2y}{dx^2}$$

$$f'(x) = (3-x)y^2$$

$$f''(x) = (3-x)2y + y^2(-1)$$

~~$$f''(1) = \frac{d^2y}{dx^2}$$~~

$$f''(1) = \frac{d^2y}{dx^2} \bigg|_{(1, -1)} = (3-1)2(-1) + (-1)^2(-1)$$

$$= (2)(-2) - 1 = -5$$

PART B

~~$$f'(x) \quad f'(1) = (3-1)$$~~

$$\frac{dy}{dx} \bigg|_{(1, -1)} = (3-1)(-1)^2 = 2$$

$$f(x) + \frac{f'(x)x}{1!} + \frac{f''(x)x^2}{2!} \quad (cx.)$$

$$-1 + 2(x-1) - \frac{5(x-1)^2}{2!}$$

Page 12

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Q5

**NO CALCULATOR ALLOWED**

Q5

Answer QUESTION 5 PARTS C and D on this page.

**PART C**

Lagrange error bound \*

$$|\text{Error}| \leq \frac{60}{3!} |1.1 - 1|^3$$

**PART D**

$$\begin{array}{c|c|c} (1, -1) & y + 1 = 2(x - 1) & (1.2, -.6) \\ (1.2, -.6) & y + .6 = \frac{81}{125}(x - 1.2) & (1.4, \end{array} \quad \frac{dy}{dx} = (3 - x)(y^2)$$

$$f(1.4) \approx \frac{81}{125} (1.4 - 1.2) - .6 \left( \frac{2}{10} + \frac{1}{10} \right)$$

$$\begin{aligned} & (3 - 1.2)(.6)^2 \\ & \left( \frac{94}{100} \right) \left( \frac{9}{100} \right) = \frac{81}{125} \\ & 2.8 \end{aligned}$$

$$\begin{aligned} f(1.4) & \approx (3 - 1.2)(-.6)^2 (1.4 - 1.2) - .6 \left( \frac{2}{10} \right) (2) = \frac{1}{10} + \frac{10}{10} = -\frac{6}{10} \\ & \approx (1.8)(-.6)^2 (.2) - .6 \end{aligned}$$

Page 13

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Q5

 NO CALCULATOR ALLOWED

Q5

Answer QUESTION 5 PARTS A and B on this page.

## PART A

$$\frac{dy}{dx} = (3-x)(y^2) \quad \frac{1}{3} + 1 = C$$

$$C = \frac{4}{3}$$

$$\int y^2 dy = \int (3-x) dx$$

$$-\frac{1}{3}y^3 = -1 + C$$

$$\frac{1}{3}y^3 = -1 + \frac{4}{3}$$

$$-\frac{1}{3}y^3 = -1 + C$$

$$\frac{1}{3}y^3 = 1 - \frac{4}{3}$$

$$\frac{1}{y^3} = 3 - 4$$

$$\frac{1}{3(-1)^3} = -1 + C$$

$$\frac{1}{y^3} = -1$$

$$f''(1) = -1$$

$$-\frac{1}{3} = -1 + C$$

$$1 = -y^3$$

$$-1 = y^3 \quad y = -1$$

?

## PART B

$$\frac{f'(x)(x-c)^n}{n!}$$

$$f(1) = -1$$

$$f'(1) =$$

$$f''(1) = -1$$

Q5

NO CALCULATOR ALLOWED

Q5

Answer QUESTION 5 PARTS C and D on this page.

## PART C

$$\frac{f(z)(x-c)^{n+1}}{(n+1)!} \rightarrow \frac{(60)(0.1)^3}{3!} \leq 0.01$$

$$\begin{array}{r} 0.1 \\ 0.1 \\ \hline 0.01 \\ 0.01 \\ \hline 0.02 \\ 0.02 \\ \hline 0.04 \\ 0.04 \\ \hline 0.08 \\ 0.08 \\ \hline 0.16 \end{array}$$

$$\frac{60(0.001)}{6}$$

$$10(0.001) \leq 0.01$$

$$0.01 \leq 0.01$$

## PART D

$$\Delta x = 0.2 \quad \frac{dy}{dx} = (3-x)(y^2)(dx)$$

$$(1, -1) \quad (3-1)((-1)^2)(0.2) = (2)(0.2) = 0.4$$

$$(1.2, -0.6) \quad (3-1.2)((-0.6)^2)(0.2) = (1.8)(0.36)(0.2)$$

$$(1.4, -0.470)$$

$$f(1.4) \approx -0.470$$

$$\begin{array}{r} 3 \\ 0.6 \\ 0.6 \\ \hline 0.36 \\ 0.36 \\ \hline 0.72 \\ 0.72 \\ \hline 1.44 \end{array}$$

$$\begin{array}{r} 1.8 \\ 0.8 \\ \hline 0.36 \\ 0.36 \\ \hline 0.72 \\ 0.72 \\ \hline 1.44 \end{array}$$

$$\begin{array}{r} 1.8 \\ 0.8 \\ \hline 0.36 \\ 0.36 \\ \hline 0.72 \\ 0.72 \\ \hline 1.44 \end{array}$$

## Question 5

**Note:** Student samples are quoted verbatim and may contain spelling and grammatical errors.

### Overview

**NEW for 2025:** The question overviews can be found in the *Chief Reader Report on Student Responses on AP Central*.

### Sample: 5A

**Score: 9 (1-1-1-1-1-1-1-1-1)**

The response earned 9 points: 3 points in part A, 2 points in part B, 2 points in part C, and 2 points in part D.

In part A the response earned **P1** with the correct application of the product rule in the expression

$-y^2 + 2(3-x)y \frac{dy}{dx}$  in line 1. The response earned **P2** with the correct application of the chain rule in the expression  $-y^2 + 2(3-x)y \frac{dy}{dx}$  in line 1. The response earned **P3** with the equation  $f''(1) = 2(2)^2(-1)^3 - (-1)^2$  in line 2.

In part B the response earned **P4** with the two correct terms,  $-1$  and  $2(x-1)$ , of the Taylor polynomial in line 1. The response earned **P5** with the third correct term,  $-\frac{9}{2}(x-1)^2$ , in line 1.

In part C the response earned **P6** with the expression  $\left| \frac{60}{3!}(1.1-1)^3 \right|$  in line 1. The response earned **P7** by connecting the expression for error,  $|f(1.1) - T_2(1.1)|$ , and 0.01 to the value of the error bound with a correct inequality.

In part D the response earned **P8**. The first step of Euler's method is shown on line 2 with the expression  $f(1.2) \approx -1 + 0.2((3-1)(-1)^2)$ . The response earned **P9** with the expression  $f(1.4) \approx -0.6 + 0.2((3-1.2)(-0.6)^2)$  on line 3.

### Sample: 5B

**Score: 6 (1-0-0-1-1-1-0-1-1)**

The response earned 6 points: 1 point in part A, 2 points in part B, 1 point in part C, and 2 points in part D.

In part A the response earned **P1** with the equation  $f''(x) = (3-x)2y + y^2(-1)$  in line 2. The response did not earn **P2**. The response does not present the chain rule factor of  $\frac{dy}{dx}$  with the first term in the product rule. The response did not earn **P3**. The response is not eligible for this point because **P2** is not earned.

In part B the response earned **P4** with the two correct terms,  $-1$  and  $2(x-1)$ , of the Taylor polynomial in line 4. The response earned **P5** with the term  $\frac{-5(x-1)^2}{2!}$ , which is consistent with the value of  $f''(1)$  imported from part A.

**Question 5 (continued)**

In part C the response earned **P6** with the expression  $\frac{60}{3!}|1.1 - 1|^3$ . The response did not earn **P7** because there is no explicit connection to 0.01.

In part D the response earned **P8**. The first step of Euler's method is shown on line 1 of the table with the presentation of the correct local linearization  $y + 1 = 2(x - 1)$  and its evaluation to produce the point  $(1.2, -0.6)$ . The response earned **P9** with the expression  $f(1.4) \approx (3 - 1.2)(-0.6)^2(1.4 - 1.2) - 0.6$  in the second to last line.

**Sample: 5C****Score: 3 (0-0-0-0-1-0-1-1)**

The response earned 3 points: 0 points in part A, 0 points in part B, 1 point in part C, and 2 points in part D.

In part A the response did not earn **P1**, **P2**, or **P3**. In an attempt at the alternate solution using separation of variables, the response produces an incorrect solution for the function  $y$  and is not eligible for any points in part A.

In part B the response did not earn **P4** or **P5** because no terms of the Taylor polynomial are presented.

In part C the response earned **P6** with the expression  $\frac{(60)(0.1)^3}{3!}$  in the middle of line 1. The response did not earn **P7** because there is no explicit connection to error or error bound.

In part D the response earned **P8**. The first step of Euler's method is started on line 2 with the presentation of the correct initial value, value of the derivative, and correct step size. It is completed with the presentation of the point  $(1.2, -0.6)$  on line 3. The response earned **P9** with the correct work on line 3 and the presentation of the answer  $f(1.4) \approx -0.470$  on line 5.